

GenAI. Music and sound gen

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March 9, 2025



LAMBDA • HSE

Introduction

- ▶ Traditional sound generation methods: analog and digital synthesizers.
- ▶ Modern methods: neural network models for music generation.
- ▶ Key technologies: MusicGen Large, Stable Audio, AudioLDM2.

MusicGen: Simple and Controllable Music Generation

Stable Audio Research

AudioLDM 2: Learning Holistic Audio Generation with Self-supervised Pretraining

Background



Mel spectrogram

The conversion from frequency f in Hz to the Mel scale is given by:

$$m(f) = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \quad (1)$$

The inverse transformation from the Mel scale back to frequency (Hz) is:

$$f(m) = 700 \cdot \left(10^{\frac{m}{2595}} - 1 \right) \quad (2)$$

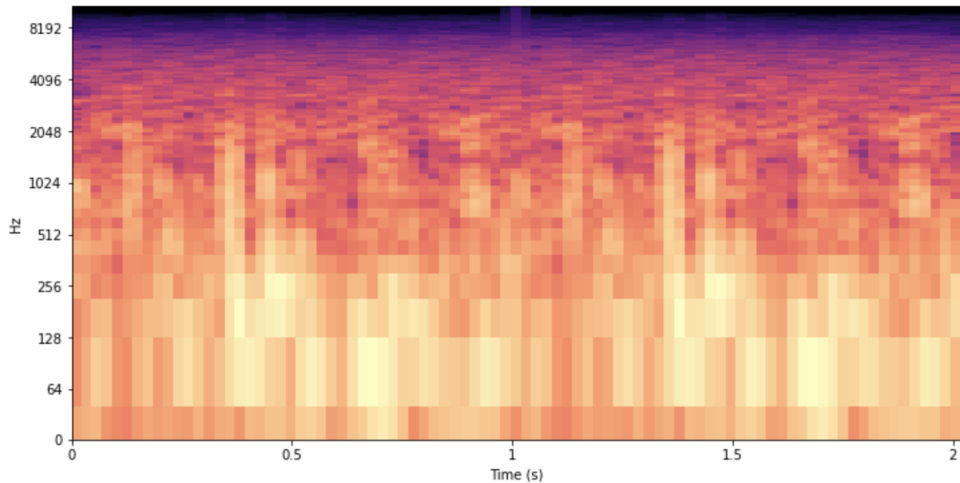
The Mel spectrogram is computed using the Short-Time Fourier Transform (STFT) as:

$$S_{\text{mel}}(m, t) = M_{m,k} \cdot |S(k, t)| \quad (3)$$

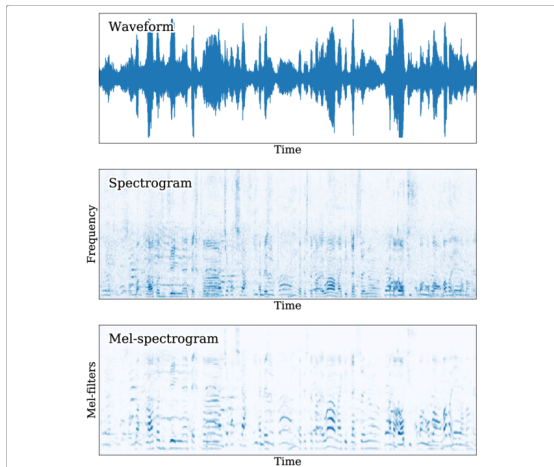
where:

- ▶ $S(k, t)$ is the STFT magnitude spectrogram,
- ▶ $M_{m,k}$ is the Mel filterbank matrix.

Mel spectrogram

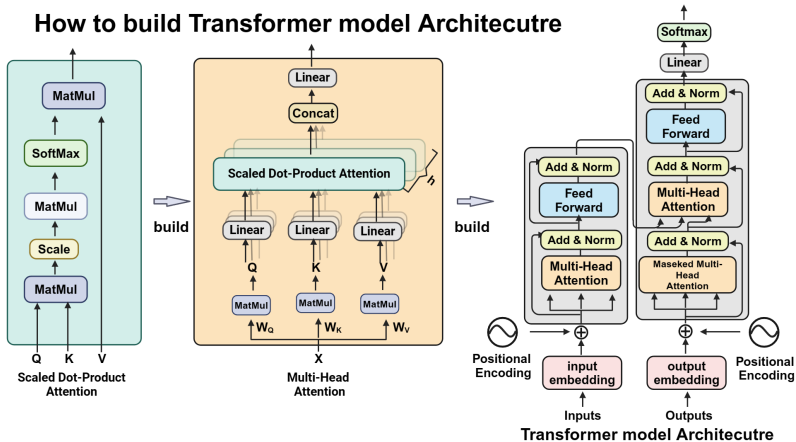


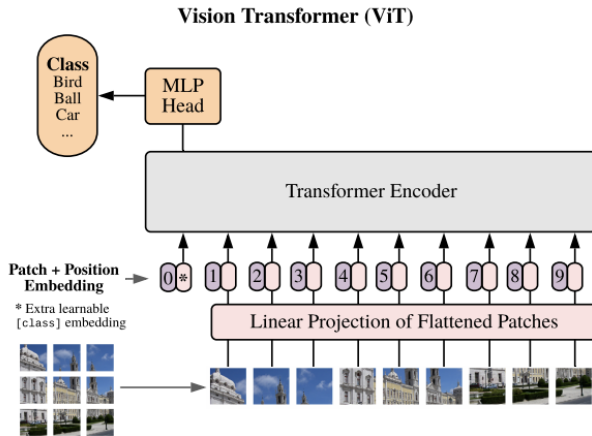
Mel spectrogram vs others



Transformer

How to build Transformer model Architecture

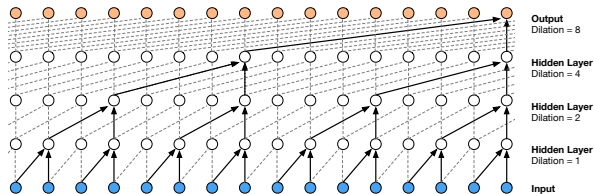




WaveNet

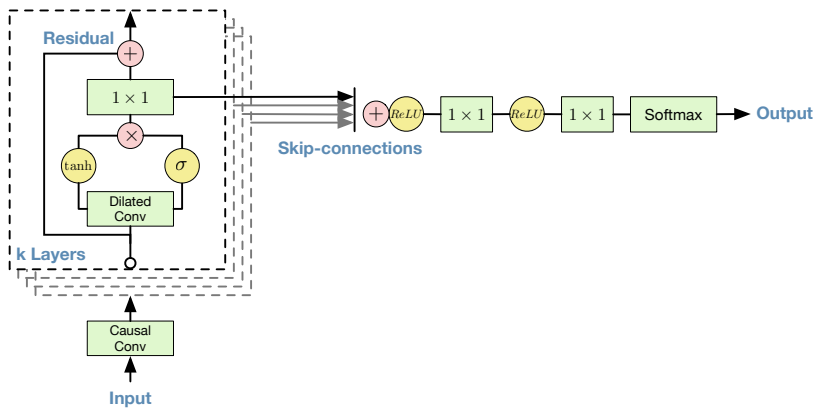


WaveNet. Dilated convolution



WaveNet

WaveNet. Dilated convolution



WaveNet

WaveNet. Summary

- ▶ Because raw audio is typically stored as a sequence of 16-bit integer values (one per timestep), a softmax layer would need to output 65,536 probabilities per timestep to model all possible values. To make this more tractable, they first apply a μ -law companding transformation (ITU-T, 1988) to the data, and then quantize it to 256 possible values:

$$f(x_t) = \text{sign}(x_t) \frac{\ln(1 + \mu|x_t|)}{\ln(1 + \mu)}$$

- ▶ the output is generated in an autoregressive format:

$$\mathbf{z} = \tanh \left(W_{f,k} * \mathbf{x} + V_{f,k}^T \mathbf{h} \right) \odot \sigma \left(W_{g,k} * \mathbf{x} + V_{g,k}^T \mathbf{h} \right)$$

$$p(\mathbf{x} \mid \mathbf{h}) = \prod_{t=1}^T p(x_t \mid x_1, \dots, x_{t-1}, \mathbf{h})$$

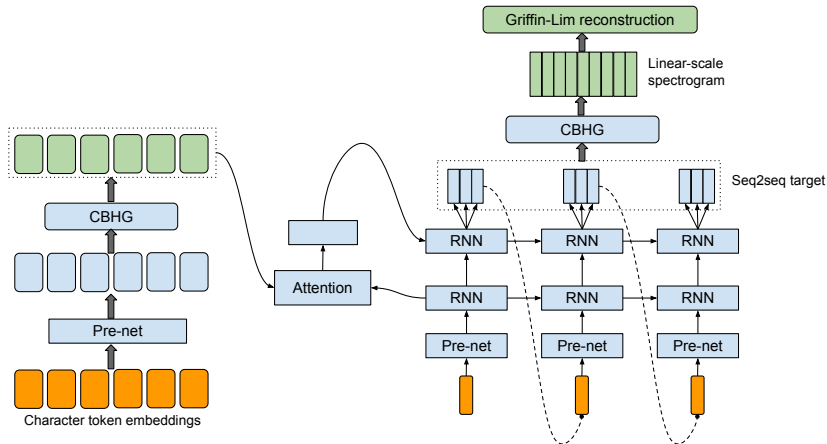
- ▶ In Parallel Wavenet, an IAF is used

WaveNet

Tacotron

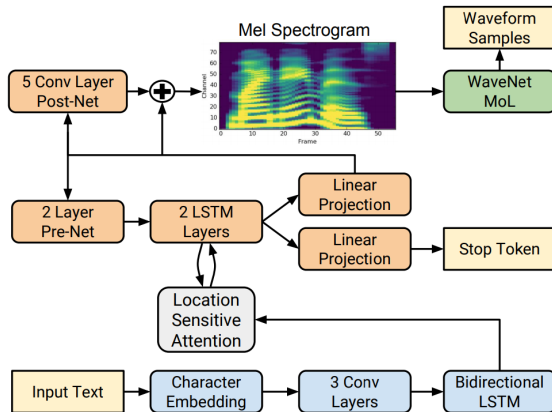


Tacotron



Tacotron

Tacotron2



Tacotron 2

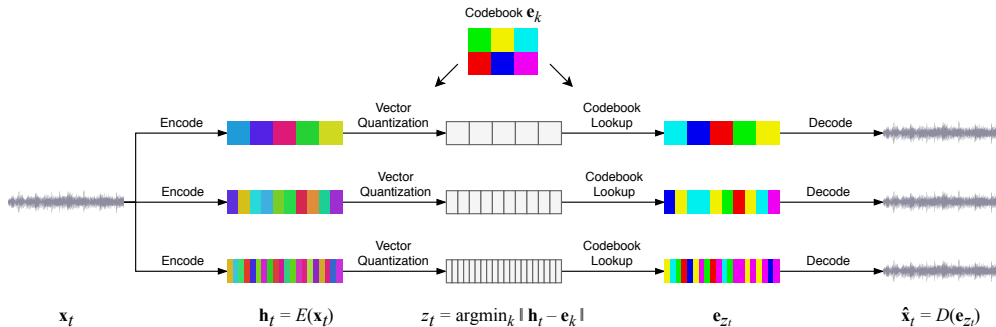
Tacotron 2: Key Improvements over Tacotron 1

- ▶ **End-to-End Speech Synthesis with Mel-Spectrograms**
 - Tacotron 1 predicted **linear spectrograms**, requiring extra post-processing.
 - Tacotron 2 predicts **mel-spectrograms**, making synthesis **smoother and more compact**.
- ▶ **Replacing the Griffin-Lim Vocoder with WaveNet**
 - Tacotron 1 used Griffin-Lim, resulting in robotic-sounding speech.
 - Tacotron 2 replaces it with a **WaveNet vocoder**, leading to **significantly improved audio quality**.
- ▶ **Improved Sequence-to-Sequence Model**
 - Tacotron 1 had **attention failures**, causing word repetition or missing words.
 - Tacotron 2 uses **location-sensitive attention**, improving text-to-speech alignment.
- ▶ **More Efficient and Simplified Architecture**
 - Tacotron 1 required complex CBHG + Bi-GRU processing.
 - Tacotron 2 simplifies the pipeline using **CNN + BiLSTM**, making training and inference **more stable and efficient**.

Jukebox (OpenAI)

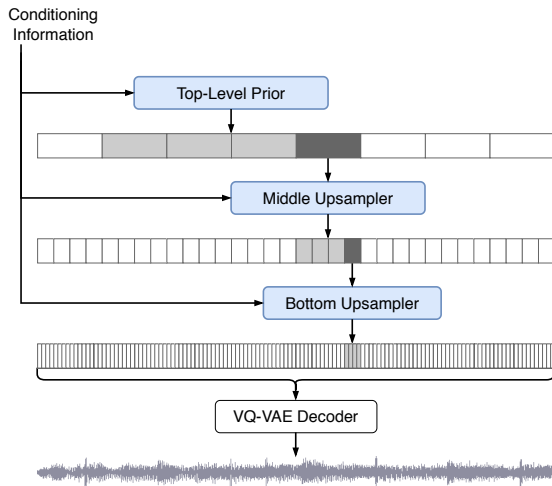


Jukebox



Jukebox

Jukebox

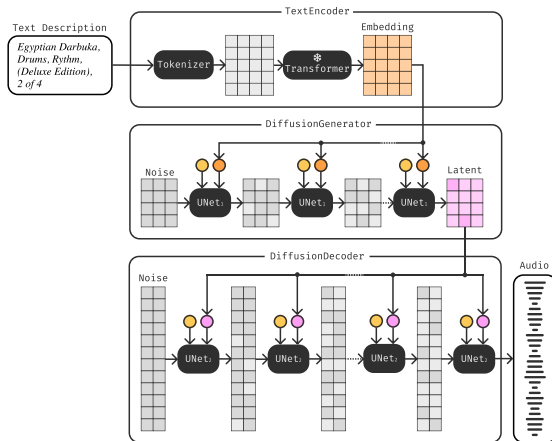


Jukebox

Moûsai

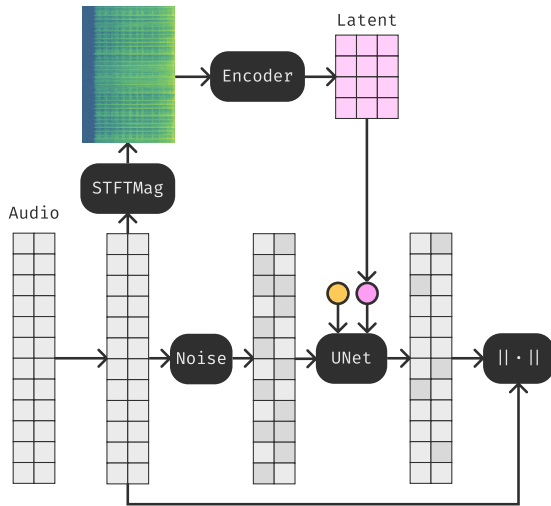


Moûsai



Moûsai

Moùsai. Diffusion Autoencoder



MusicGen Large (Meta AI)

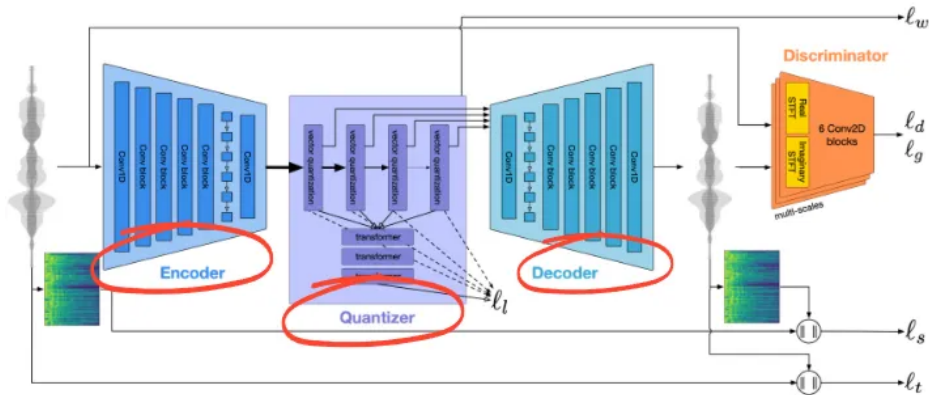


MusicGen Large

- ▶ Developed by Meta AI.
- ▶ Uses transformers and vector quantization (VQVAE) for music generation.
- ▶ Enables the creation of audio tracks from text descriptions.
- ▶ Applications: automatic composition, soundtrack generation.

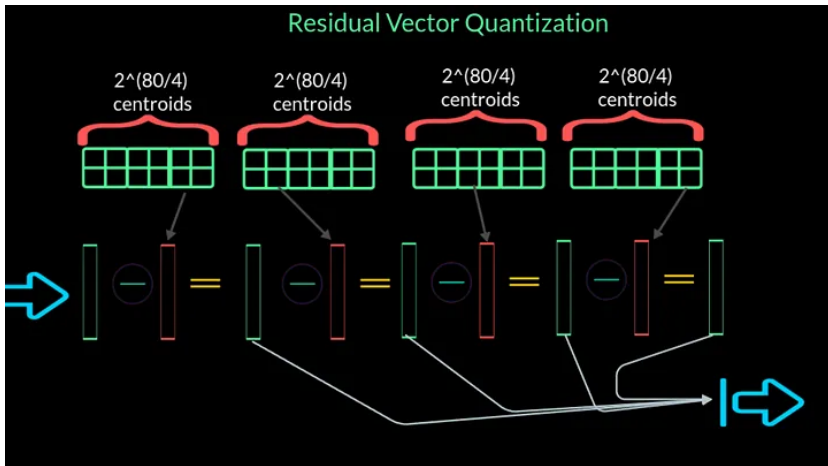
MusicGen: Simple and Controllable Music Generation

MusicGen Large



MusicGen: Simple and Controllable Music Generation

MusicGen Large. Residual quantization



MusicGen: Simple and Controllable Music Generation

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Stable Audio



Stable Audio (Stability AI)

- ▶ Developed by Stability AI.
- ▶ Generates sound with high accuracy, supporting diverse styles.
- ▶ Based on diffusion models, similar to Stable Diffusion in images.
- ▶ Applications: background music generation, sound effects.

Stable Audio Research

Stable Audio

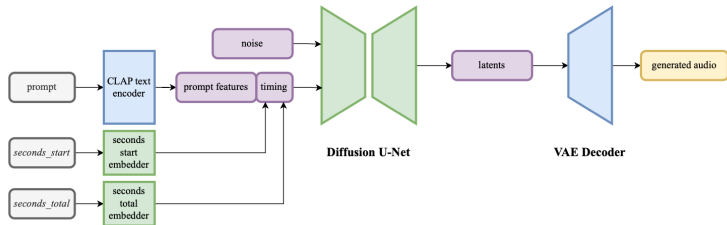


Figure 1. Stable Audio. Blue: frozen pre-trained models. Green: parameters learnt during diffusion training. Purple: signals of interest.



****Two-stage training process**:**

1. Train ****VAE**** (Descript Audio Codec with Snake activation).
2. Train ****diffusion model**** on compressed latent representations.

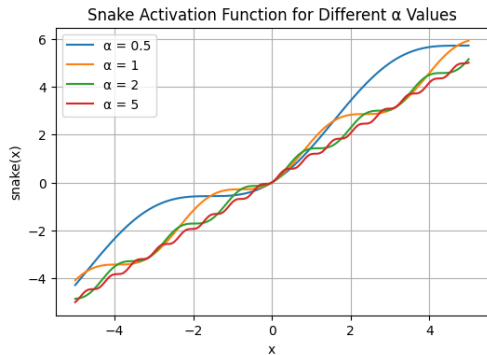
Stable Audio Research

Stable Audio

- ▶ Unlike previous work, **Stable Audio** uses **latent diffusion**, meaning diffusion happens in the **latent space** rather than tokenized representations.
- ▶ The latent space is **continuous**, so instead of **VQ-VAE**, a standard **VAE** is used.
- ▶ Specifically, **The Descript Audio Codec** is utilized as the audio compression model.
- ▶ Authors claim that when combined with **Snake activation**, this codec performs better than **RVQ** from the MusicGen paper.
- ▶ Possible reasons for this improvement:
 - **Error accumulation** at each **Residual Quantization** stage.
 - **RVQ's inability** to capture complex dependencies in audio sequences.
- ▶ The diffusion process is trained **directly on the latent space**.
- ▶ **Text-based CLAP embeddings** and **track duration metadata** are injected via **classifier-free guidance** during diffusion.

Stable Audio Research

Stable Audio: Snake Activation



Snake Activation Function

Snake activation

The Snake activation function is defined as:

$$\text{Snake}(x, \alpha) = x + \frac{1}{\alpha} \sin^2(\alpha x) \quad (4)$$

where:

- ▶ x is the input value,
- ▶ α is a learnable scaling parameter.

AudioLDM2

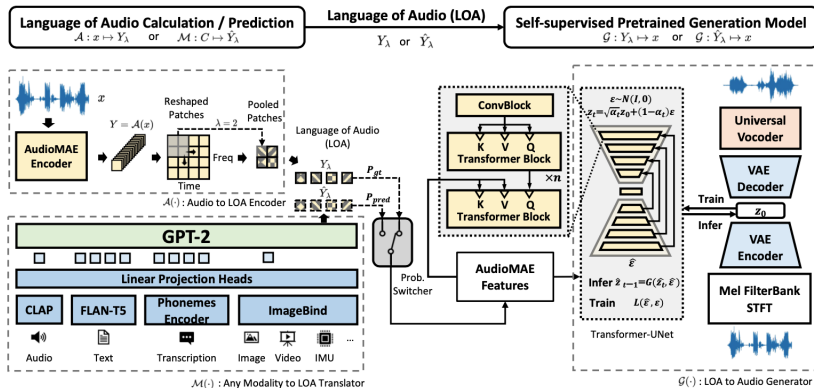


AudioLDM2 Music

- ▶ Second version of AudioLDM with improved sound and music generation.
- ▶ Uses latent diffusion modeling (LDM).
- ▶ Supports text and audio inputs for generating new sounds.
- ▶ Applications: music production, audio file adaptation.

AudioLDM 2: Learning Holistic Audio Generation with Self-supervised Pretraining

AudioLDM2



AudioLDM2

1. **AudioMAE Pretraining (Denoising Autoencoder)**
 - Works on **Mel-spectrograms**.
 - Encoder-decoder architecture inspired by **ViT (Vision Transformer)**.
 - Spectrogram is **split into patches**, and the model reconstructs missing patches (**Masked Language Modeling**, similar to BERT).
 - Ensures the **latent space (LOA)** contains all necessary semantics.
2. **Metadata Adapter (GPT-2)**
 - Maps **captions and metadata** to the same **latent LOA space** as the audio.
 - Enables **text-to-audio conditioning**.
3. **Latent Diffusion Model**
 - Uses a **Transformer-UNet** instead of a standard UNet (**similar to Diffusion Transformer, DiT**).
 - Task: Reconstruct the **original latent representation** of a song from the **adapter-generated latent**.

Conclusion

- ▶ Modern AI models enable music creation based on text descriptions.
- ▶ MusicGen Large, Stable Audio, and AudioLDM2 are cutting-edge audio generation technologies.
- ▶ Future perspectives: improved synthesis quality, sound personalization, user adaptation.

Thank you for your attention!



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